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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

First Semester, September - October, 2025

Paper Code: BS-111**Subject: Applied Mathematics-I****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1.(a)	Find $\frac{dy}{dx}$ using partial differentiation, if $e^y \sin x = x + xy$.	2.5	CO 1
1.(b)	Apply Leibnitz rule to evaluate: $\frac{d}{dt} \int_0^1 (2x + t^3)^2 dx$.	2.5	CO 1
1.(c)	Determine whether the following differential equation is exact: $[1 - \frac{x}{(x+y)^2}] dy + [\frac{y}{(x+y)^2} - 1] dx = 0$	3	CO 2
1.(d)	Show that the particular solution of the differential equation: $(x^2 + 1) \frac{dy}{dx} + (y^2 + 1) = 0$, $y(0) = 1$ is $y = (1 - x) / (1 + x)$.	2	CO 2
2.(a)	If $\mathbf{V} = f(2x - 3y, 3y - 4z, 4z - 2x)$, compute the value of $6V_x + 4V_y + 3V_z$.	5	CO 1
2.(b)	If $y_1 = (x_2 x_3) / x_1$, $y_2 = (x_3 x_1) / x_2$ and $y_3 = (x_1 x_2) / x_3$, show that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4.	5	CO 1
3.(a)	Find the local maximum and minimum values and saddle points of: $f(x, y) = x^4 + y^4 - 4xy + 1$.	5	CO 1
3.(b)	If $u = \sin^{-1}(x/y) + \tan^{-1}(y/x)$, show that $x(\partial u / \partial x) + y(\partial u / \partial y) = 0$.	5	CO 1
4.(a)	Find the solution of the initial-value problem: $x^2 \frac{dy}{dx} + xy = 1$, $x > 0$, $y(1) = 2$.	5	CO 2
4.(b)	Solve the following Bernoulli differential equation: $\cos x dy = y(\sin x - y) dx$.	5	CO 2

**** End of Question Paper ****